

Labwork 3: Hello, CUDA!

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1 Method

I read input.jpg with imread. The image is 512 pixels wide and 600 pixels high. I reshape it to (512 * 600, 3) and calculate the gray value of each pixel:

$$\text{gray} = (R + G + B) / 3$$

On the CPU, I use a for loop. On the GPU, each thread handles one pixel. Its index is threadIdx.x + blockIdx.x * blockDim.x. I check that the index is smaller than the number of pixels. I convert R, G and B to integers before adding them to avoid overflow.

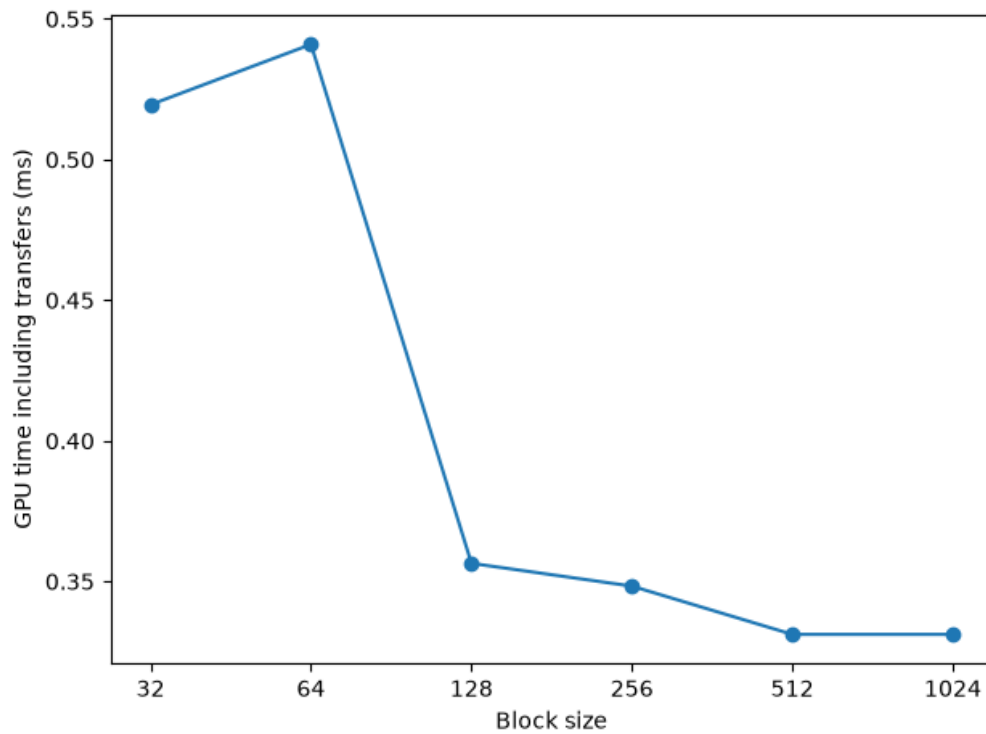
2 Results

I use time.time() to measure the CPU and GPU times. I run the kernel once before timing so compilation is not included. I then measure each block size once. GPU time includes copying the image to the GPU and copying the result back. I divide CPU time by GPU time to get the speedup.

```
CPU time: 120.655 ms
Block size: 32
GPU time: 0.52 ms
Speedup: 232.25
Block size: 64
GPU time: 0.541 ms
Speedup: 223.03
Block size: 128
GPU time: 0.356 ms
Speedup: 338.5
Block size: 256
GPU time: 0.348 ms
Speedup: 346.38
Block size: 512
GPU time: 0.331 ms
Speedup: 364.34
Block size: 1024
GPU time: 0.331 ms
Speedup: 364.34
```

3 Comments

The CPU and GPU give the same grayscale image. The CPU takes 120.655 ms. Block sizes 512 and 1024 both take about 0.331 ms, giving a speedup of about 364. A larger block is not always faster: block size 64 takes slightly longer than 32. I only measured each size once, so these small differences may change in another run.



From left to right: input image, CPU result, GPU result.